

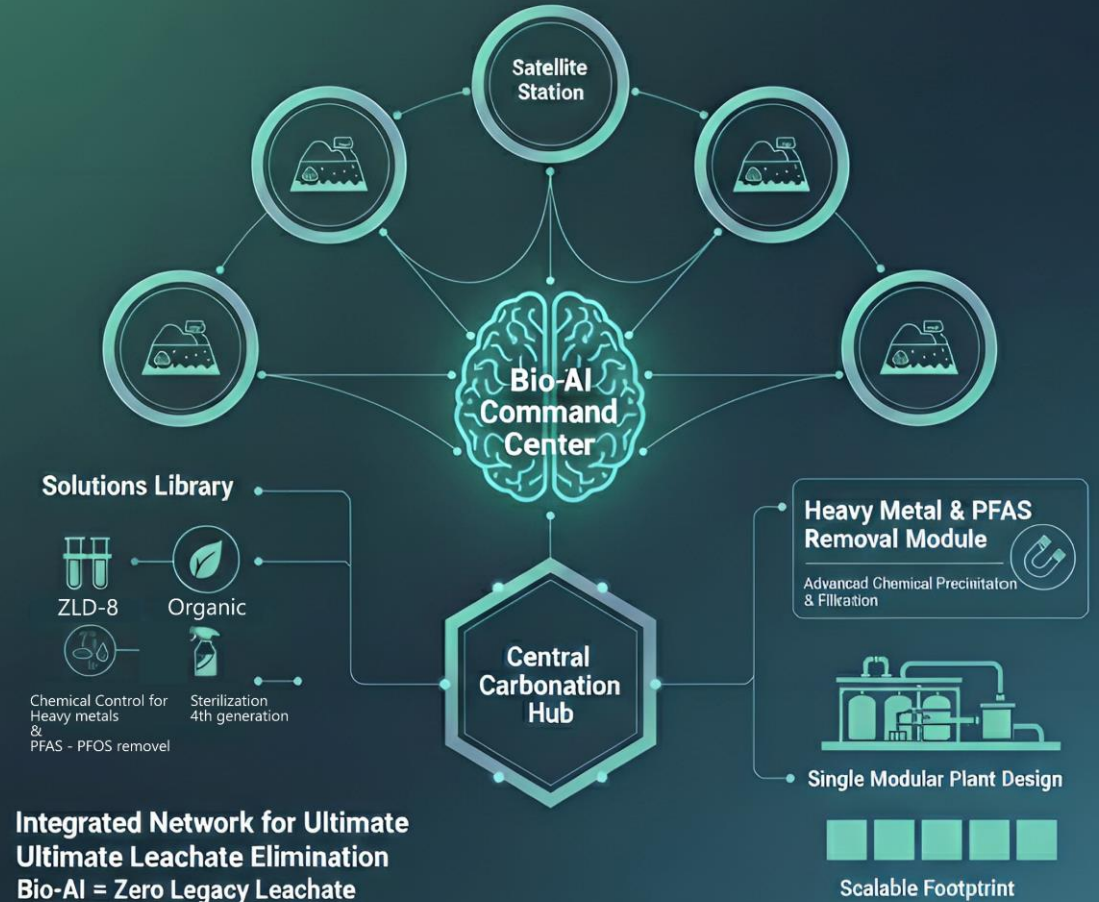
A Revolutionary Chemical-Physical Protocol for
Permanent Sludge Elimination

The ZLD-8 Solution



BIOWATER: THE GLOBAL WASTE REVOLUTION

POWERED BY BIO-AI



IP Security

OpEx Savings

Zero Emission

Scalability

Real-Time Control



This document presents a complete ZLD-8 treatment system through a full Technology Transfer program covering :

- Process,
- Know-How,
- OEM Supply Chain.

- The Company: BAYWATER SARL
- Based : Casablanca (Morocco)
- R&D Alliance Technology: Lebanese / Japanese / Chinese

➔ Immediate and measurable results; not theoretical promises.

Context : The Structural Failure of Conventional Zero Liquid Discharge (ZLD) Systems

The Problem

Traditional ZLD systems (RO, evaporation, thermal) are designed to purify water.

The Structural Flaw

These systems generate concentrated brine and sludge that are returned to landfills, creating a continuous “**Toxic Loop**” and biological reactivation.



The Reality

Aged landfill leachate is not wastewater but a **high-density liquid sludge**, resistant to conventional biological treatment due to refractory organic matter (humic acids).



Leachate : The "**Liquid Sludge**" Paradox



The Trap: Engineers often treat leachate as Industrial Wastewater. This is a structural mistake.



The Reality: Industrial wastewater is "Water-Based" (Water that became contaminated).



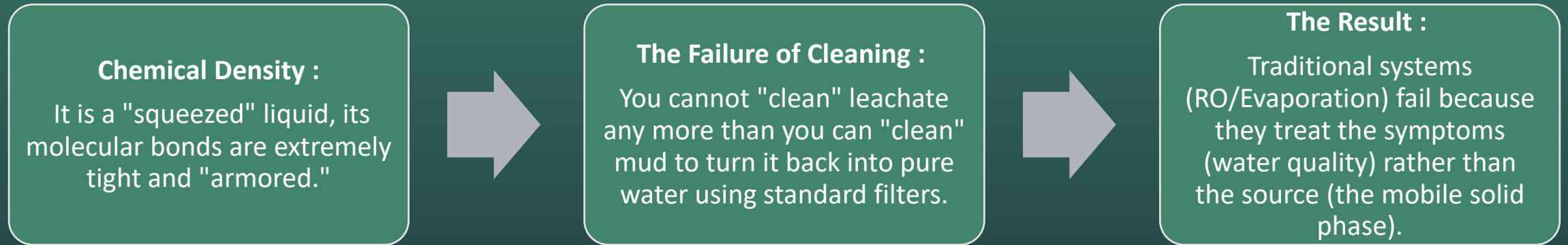
The Truth: Leachate is "Solid-Based" (A liquid created by the biochemical squeezing and extraction of solids).



The Definition: Leachate is not water; it is a Highly Concentrated Liquid Sludge that merely “mimics” the appearance of water.

Leachate : The "**Liquid Sludge**" Paradox

Why it Defeats Conventional Treatment ?



Conclusions :

#1 : Leachate is a solid-phase monster disguised as a liquid.

#2 : Conventional systems try to filter the liquid; ZLD-8 arrests the solid and terminates its environmental cycle.

The 5 Core Pillars of **ZLD-8** Technology by Baywater.

1 - Identity & Vision:

ZLD-8: Transforming Environmental Issues into Industrial Value.

2 - The Scientific Correction:

The Liquid Sludge Paradigm: Leachate is Liquefied Waste, Not Contaminated Water.

3 - Treatment Philosophy:

Water as a Strategic Carrier: Utilizing Molecular Expansion for Solid-Phase Extraction.

4 - Process & Results:

The 25-Minute Transformation: From Toxic Liquid to Inert Powder and Secure Resin Blocks.

5 - Security & Bio-AI:

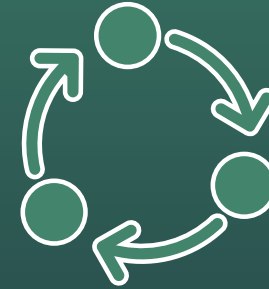
Safeguarding Intellectual Property via Bio-AI & Ensuring Product Safety through **TCLP** Standards.

Converting Environmental Liabilities into Industrial Assets

The Future of Zero-Old-Leachate Technology Outline Presentation

The ZLD-8 Solution is a revolutionary system designed for complete leachate elimination.

- Containerized and compact design
- Focused on efficient sludge management
- Zero-liquid-discharge technology
- Offered as a full acquisition, ensuring seamless integration
- Ideal for sustainable waste treatment practices

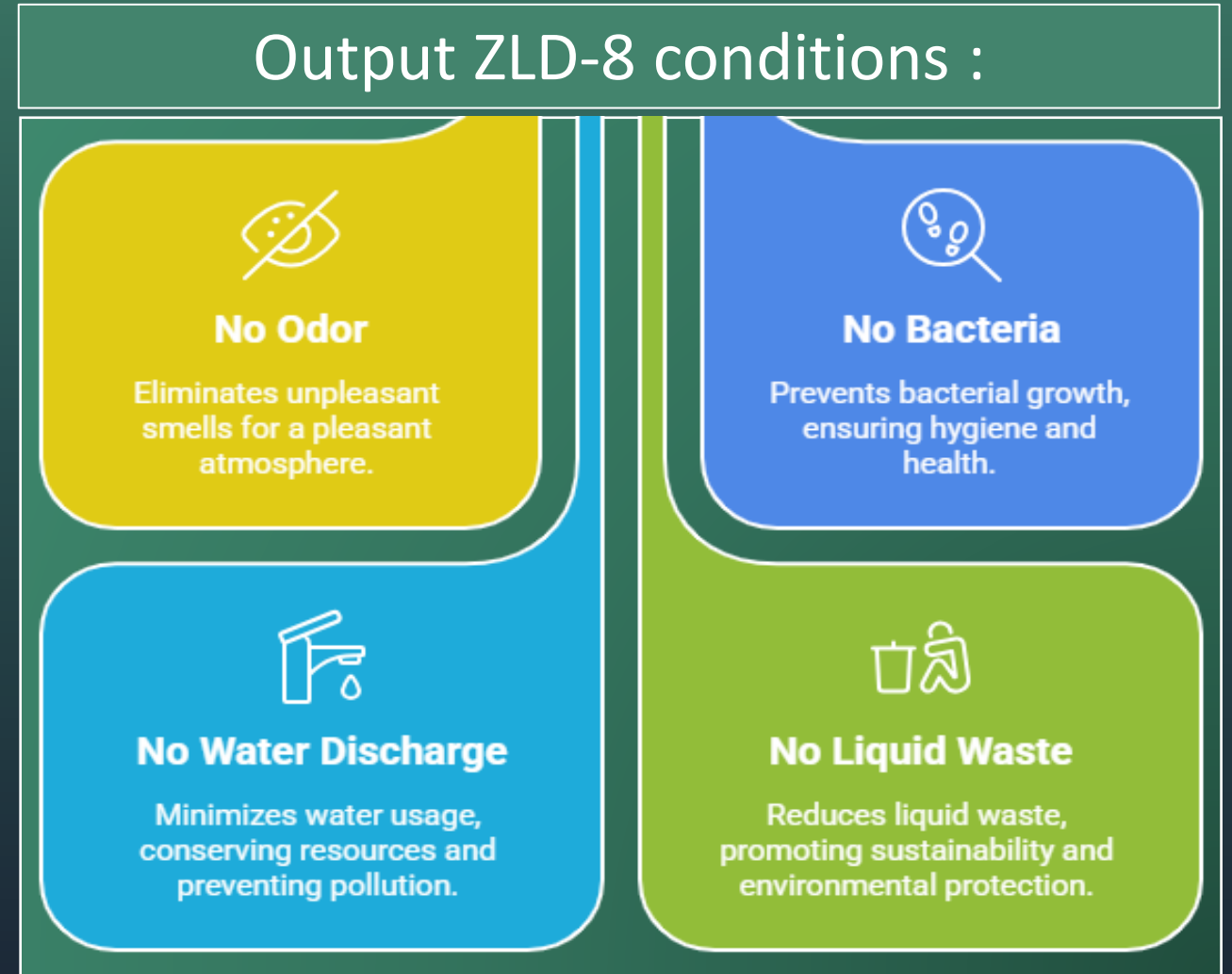


The ZLD-8 Solution operates on a fundamental principle:

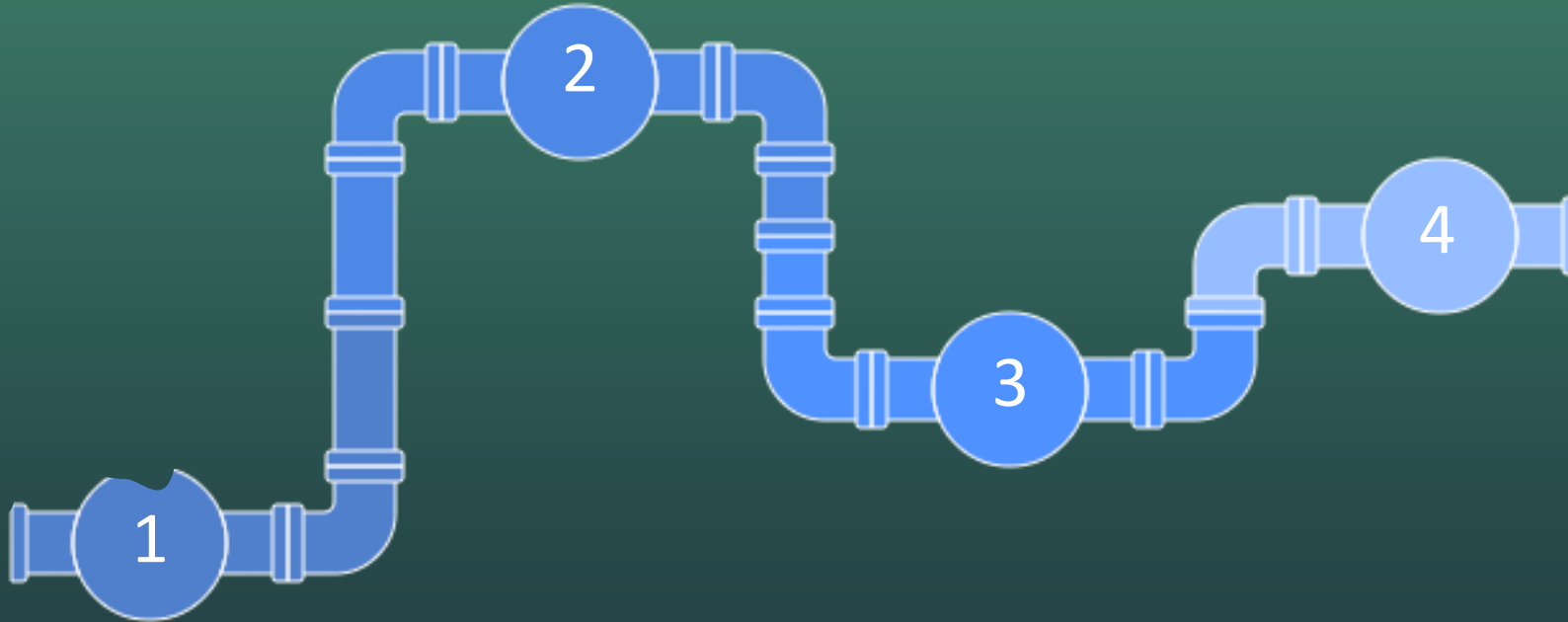
- Converts sludge to dry powder
- Utilizes a chemical–physical sequence
- Features a closed-loop water system
- Ensures no effluent discharge
- Focuses on zero-liquid discharge technology

The ZLD-8 Paradigm Shift by Baywater

- **Core Concept:** We do not treat water; we extract and convert the solid fraction (sludge).
- **Carrier Medium:** Water is merely a carrier that circulates in a fully closed loop; no liquid discharge leaves the ZLD-8 system.
- **Speed & Efficiency:** The conversion process is completed within 20–25 minutes.

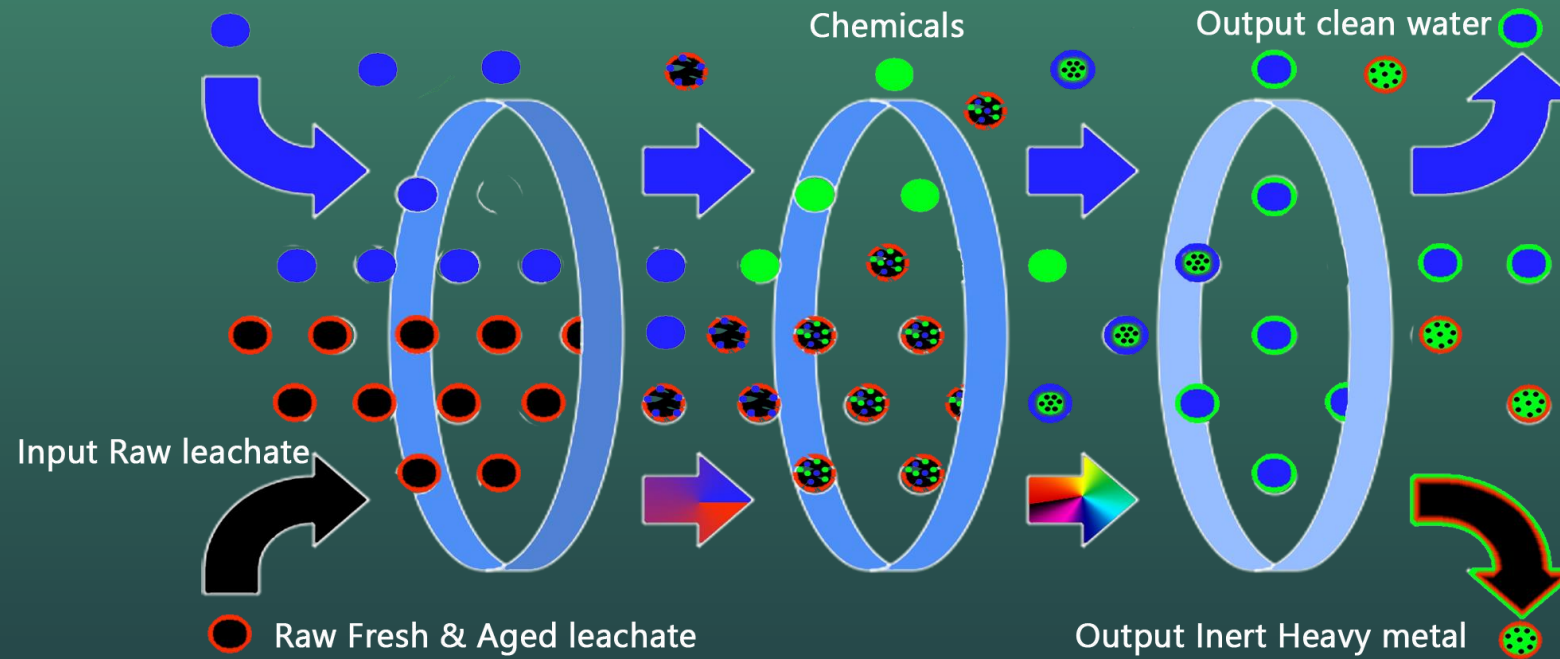


ZLD-8 Technical Workflow (Process logic)



- Step 1: **Rapid Extraction:** Immediate separation of the sludge fraction from the carrier medium.
- Step 2: **Stabilization:** Deep chemical-physical neutralization of heavy metals and pollutants.
- Step 3: **Dewatering:** High-efficiency mechanical compression (using advanced Multi-Disk Screw Press technology).
- Step 4: **Conversion:** Transformation of sludge into a fine, dry, and biologically inert industrial powder.

The Water Strategic Role



First Role : Expand Matrix

Water creates space for chemical penetration

Second Role : Act as a Carrier

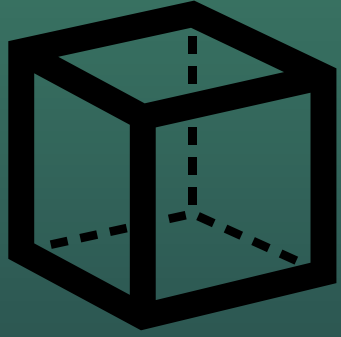
Water transports pollutants to chemical "striking" zone.

Third Role : Enable Extraction

Water allows ZLD-8 chemicals to capture "solids"

The role of water in landfill leachate treatment differs fundamentally from its role in industrial wastewater treatment. In industrial wastewater systems, water is treated through filtration processes aimed at removing dissolved and suspended contaminants. In contrast, leachate treatment requires the solidification and concentration of the waste matrix, followed by the extraction of the pollutant-bearing solid fraction.

ZLD-8 Treatment System Mass Balance



1 m³ Raw/Fresh & Aged Leachate



25 to 200 kg stabilized sludge cake



7 to 20 kg Indissoluble dry powder

From Waste to Industrial Valorization

Final Product Conditions: A stable, odorless, non-leachable powder indissoluble.

What are the valorization applications of the final product?



- **Advanced Materials:** Mixed with Resin to form decorative blocks or architectural panels.
- **Construction:** Additives for cement, mortar, and bricks.
- **Roadworks:** High-resistance fillers for Asphalt & Bitumen

The final product will pass **TCLP / EN12457** tests, ensuring the material is non-hazardous.

And more Advantages

Heavy Metal Neutralization & Safety Standards

The ZLD-8 Solution treats High concentrations of PFAS, PFOS, Heavy Metals (Lead, Mercury, Chromium, Arsenic...), and toxic mineral loads.



Molecular Stabilization:

Advanced physical-chemical binding that traps heavy metals within the sludge matrix.



Solidification/Stabilization

(S/S):

Converting hazardous ions into non-leachable, inert mineral forms.



Compliance:

Final products pass TCLP (Toxicity Characteristic Leaching Procedure) and EN12457 tests.

Treatment Output : The material is classified as Non-Hazardous under EU/US standards, allowing for safe industrial reuse or landfilling.

And more Advantages

Bio-AI: The Centralized Digital Brain

ZLD-8 system is a revolutionary operational model using the **Bio-AI** Command Center.

Key Functions:

- **Autonomous Dosing:** IoT sensors control precise chemical injections across multiple satellite sites.
- **Predictive Analytics:** Real-time monitoring to prevent operational errors and ensure 100% success rate.
- **Black Box Protection:** The core chemical formulations and molecular sequences remain encrypted within the Bio-AI core.

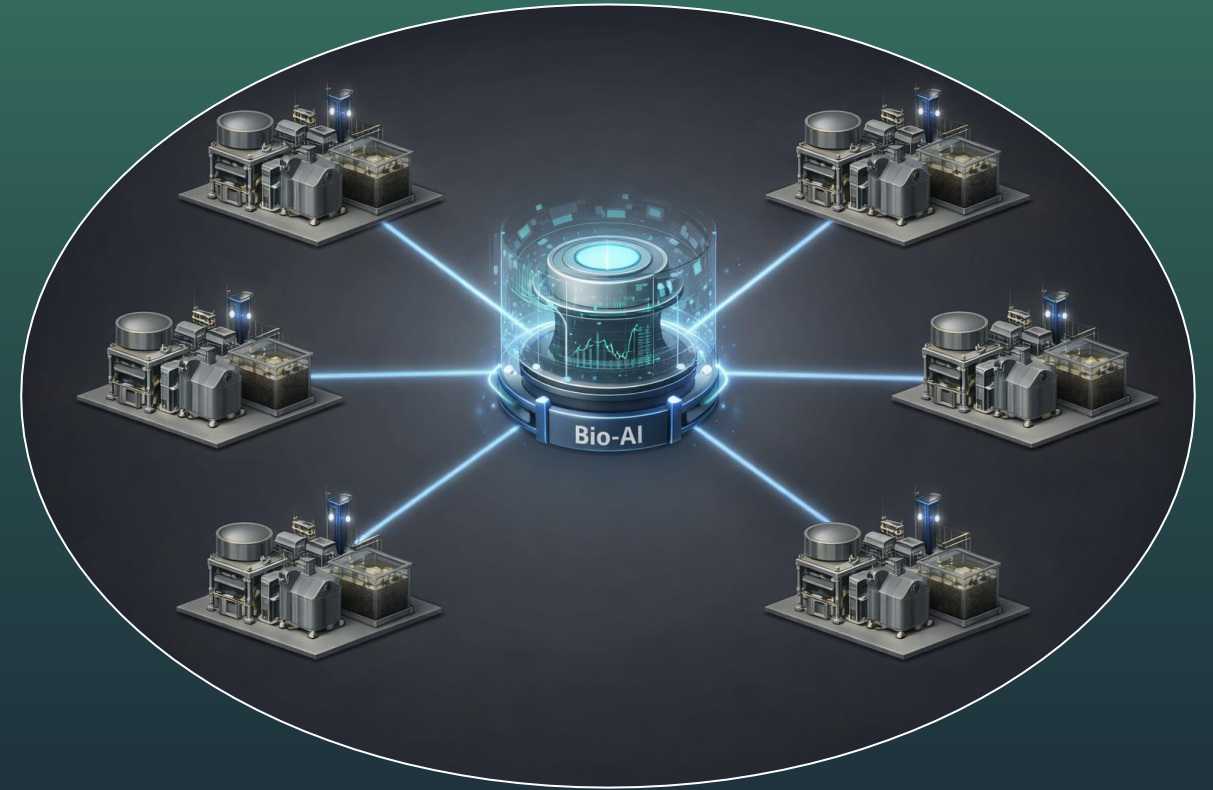


The Benefit: Local operators manage the hardware, while the AI manages the "Proprietary Recipe," ensuring total Intellectual Property (IP) security.

And more Advantages

Scalability & Strategic Architecture (The Hub Model)

- **Satellite Stations:** Multiple local stations (e.g., 5 Landfill sites) focus on leachate treatment and powder production.
- **Central Hub:** A centralized facility for carbonization and Bio-AI operations.
- **Seamless Expansion:** Easily add a 6th or 10th site by simply connecting a new satellite station to the existing Bio-AI network.



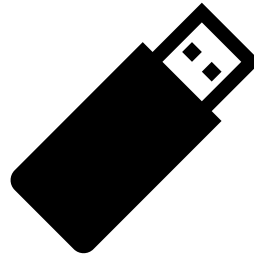
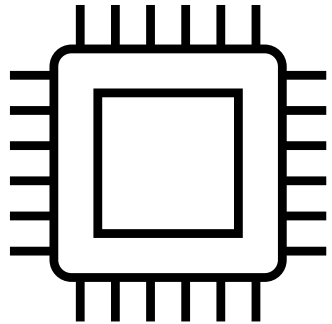
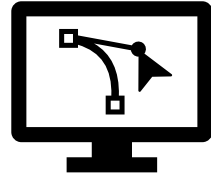
And more Advantages Efficiency

ZLD OPEX :



- **ZLD-8 System** → **25 \$/m3 (0.095 \$/gal (US))**
 - **Classical ZLD Systems** → **36 \$/m3 (0.13 \$/gal (US))**
-
- **ZLD-8** Eliminates the need for complex units at every site.
 - A centralized expert system managing by expert team, can be powered by AI, manages and optimizes chemical dosing and operations across all satellite treatment stations 24/7.

Operational Licensing & Digital Security



- **Anti-Replication:** The technology cannot be replicated or modified without a valid, verified license from the Bio-AI command.
- **Training:** Full technical replication and hands-on training provided within 15-30 days of full payment

- **Digital Key Authorization:** Access to the system is governed by a secure digital license.
- **Master Key Control:** The technology owner (Buyer) retains control over regional expansions.

Technology Transfer & Intellectual Property Protection

including

What is for Sale

Full Technology Transfer :



- **Process:** The engineered operational sequence and technical workflow of the technology.
- **Know-How:** The proprietary operational expertise, dosing strategies, and practical methods required to operate the process effectively.
- **OEM Supply Chain:** Certified manufacturing partners supplying equipment built to proprietary technical specifications.

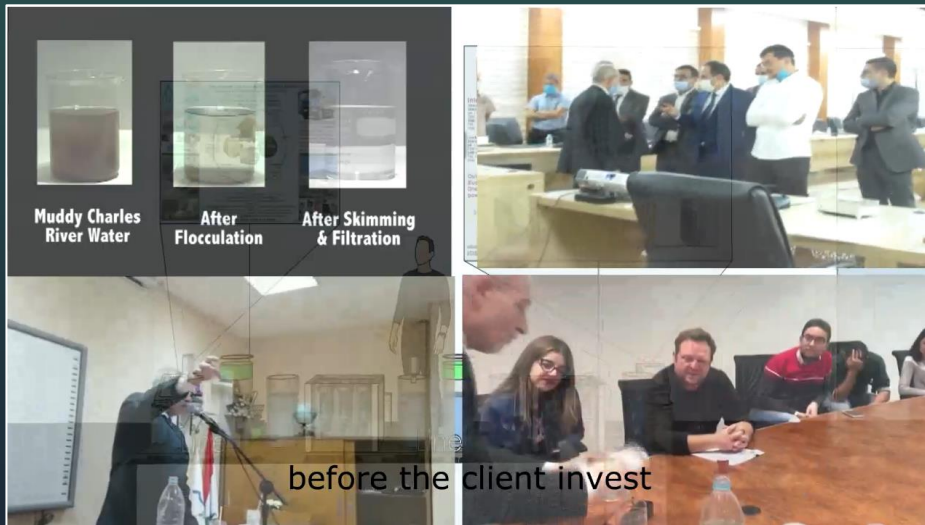
Protection Framework :



- **NDA:** Mandatory before technical disclosure.
- **Initial Payment:** 30% due immediately after successful live demo and sample handover.
- **Full Exclusivity:** Available per territory/country.
- **Manufacturer:** 15+ years of experience with 4500+ global installations 60+ national patents.
- **Chemical manufacturer sources.**
- **Media manufacturer.**

We protect IP through secrecy and immediate results ➔ Live Demonstration

The LD-Proof (Live Demonstration Model)



Zero Investment Risk

- Our live demonstrations replicate the operational sequence of the treatment system, allowing step-by-step verification of the treatment process.
- Using the client's own sample, treatment performance is validated **before any investment decision**, eliminating technical uncertainty.

Does not rely on:

- Water purification
- Evaporation
- Boiling
- Concentration
- Thermal ZLD systems

This technology is not a water-treatment solution

Goal and challenge ZERO-Old Leachate (ZLD8)

Chemical Treatment Containerized
Advanced Lebanese / Japanese / Chinese Technology
Mobile (ZLD8) Zero-Liquid-Discharge Leachate Treatment

Bio-AI: The Digital Brain (Powered by ZLD8 Logic)

Technical Overview

10 m³/h Raw leachate per line	Various raw leachate
40-m³/h Industrial wastewater	Fresh and old leachate
160m³/8 hours - two lines (Leachate)	Odorless&No bacterial
Sludge drying & Carbonization	10 tons/D
(8-10 per batch) > No output water	25kg~150kg/1m³ sludge cake
Discharge (ZLD8)inert heavy metals&PFAS	7kg~20kg/1m³ final powder

Operational & Financial Data

CapEx: 6M \$ Station containerized	OpEx: >25\$/m³ All-in, Including chemical cost	Powder resale potential 30~100 \$/ton OR Any achievable market price
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Features & Advantages

- Mobile Expandable (Fully relocatable station)
- Odour & bacterial free (Built-in Sanitizing & odour-control for landfill leachate and industrial sludge)
- Refundable

Manufacturer's Background

- +4500 Successful installation
- +60 National patent
- +100 Cooperating enterprises

Live demonstrations available (projection-ready)

Complete Treatment Sequence

Raw Leachate	Flocculation	Coagulation	Filtration
Clean final water	Sludge cake	Dry InsolublePowder	Fine soft Dry InsolublePowder

Station dimension 80m x 100m

ZLD-8 Technology Zeo-Old-Leachate

Station process

- Two lines
- Each line contain three 40ft container and one dryer.
- Capacity for each line 40m³/h Sewage/10m³/h raw leachate.
- Dryer 300kg/h
- Carbonization
- Total raw leachate treatment per 8 hours 160m³ per station.
- 1m³ raw leachate = 25kg sludge cake = 7kg powder
- Sludge cake: No odor / No bacteria / Heavy metal chelating.
- Sludge powder: No odor / no bacteria / Heavy metal chelating.
- Carbonization: inert powder PFAS / Heavy metal / Low weight.
- Resin fabrication: The laboratory determines the path.

Potential Uses of the Final Powder (Insoluble) The dried and insoluble powder produced after treatment can be integrated into various industries:

- Building & Construction Materials Additive in concrete, cement, aggregates Replacement Suitable for mortar and tiles.
- Asphalt & Bitumen Filler material improving moisture resistance & cracking performance.
- Plastics & Resins Mixed with resin to create durable, low-cost composite materials Excellent for bricks, blocks, pavers, and molded industrial pieces Market Qualification.

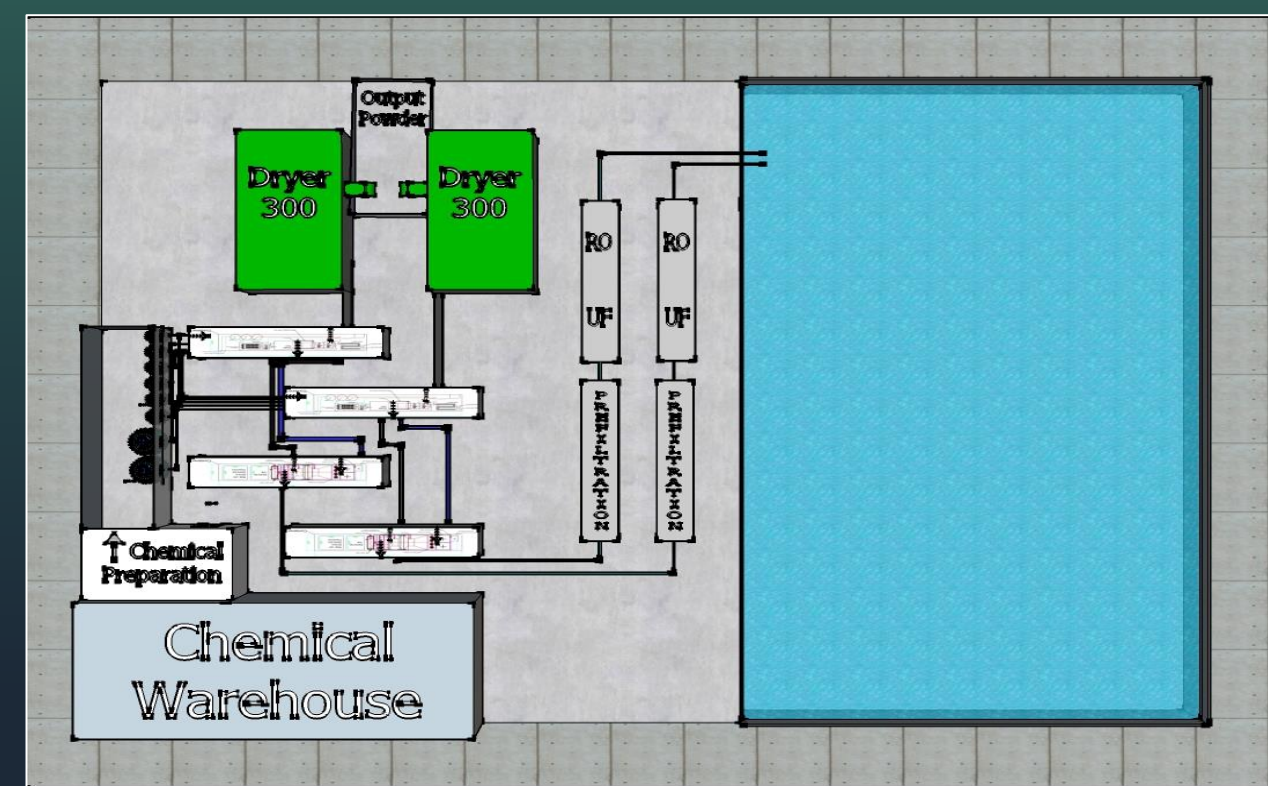
Requirements Consistent particle size Density, purity, moisture stability Basic.

certification for building/plastics applications Environmental & Economic Value Converts a dangerous waste into a commercial material No discharge □ Zero pollution □ Environmentally safe Powder is stable, dry, odourless, insoluble.

Easy to transport, store, and reuseReduces landfill costs Provides a new income stream for the buyer.

Study : Morocco Case

- Old Stock Leachate : 420,000 m³ ≈ 111 Million Gallons (US)
- Leachate Daily Flow : 650 m³ ≈ 174,000 Gallons (US)
- Station dimension : 100m x 200m
- Two lines Treatment Station :
- **Raw leachate:** 20 m³/h → 160 m³/8hours
- **Industrial Wastewater:** 40 m³/h → 320 m³/8hours
- Two Dryers : 600 kg/h



ZLD-8

Eliminates

"Old Ponds"

problem forever